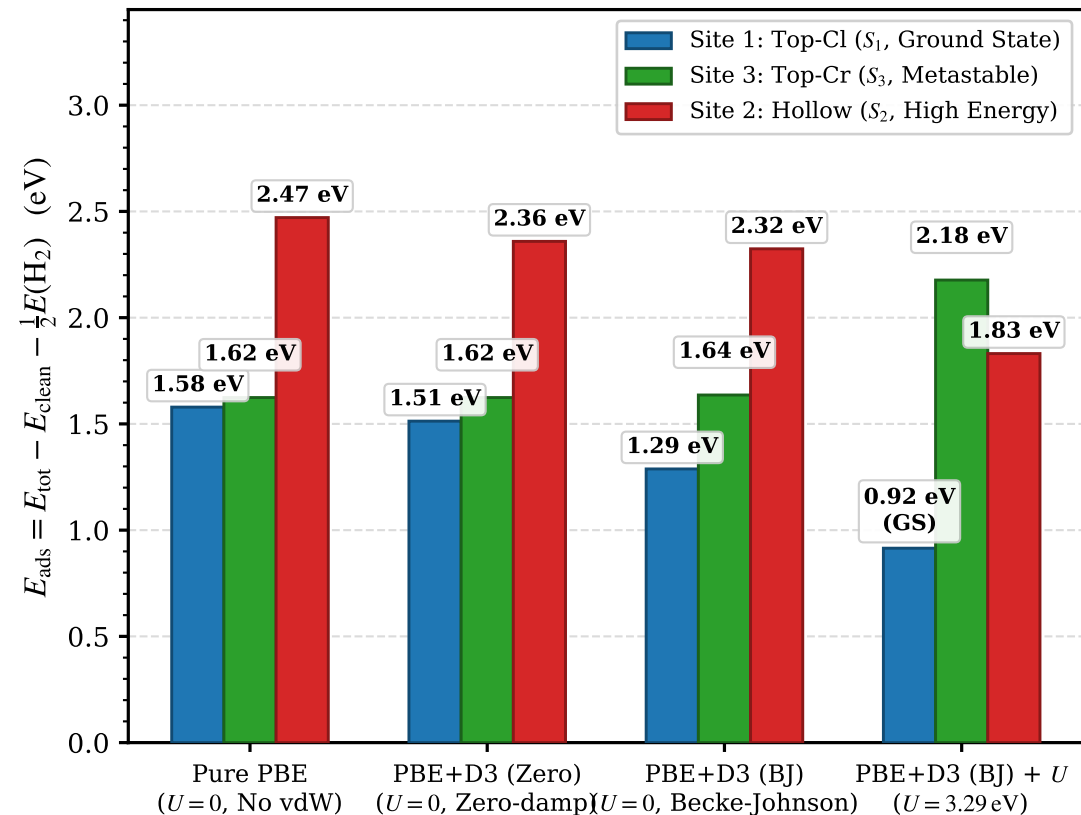
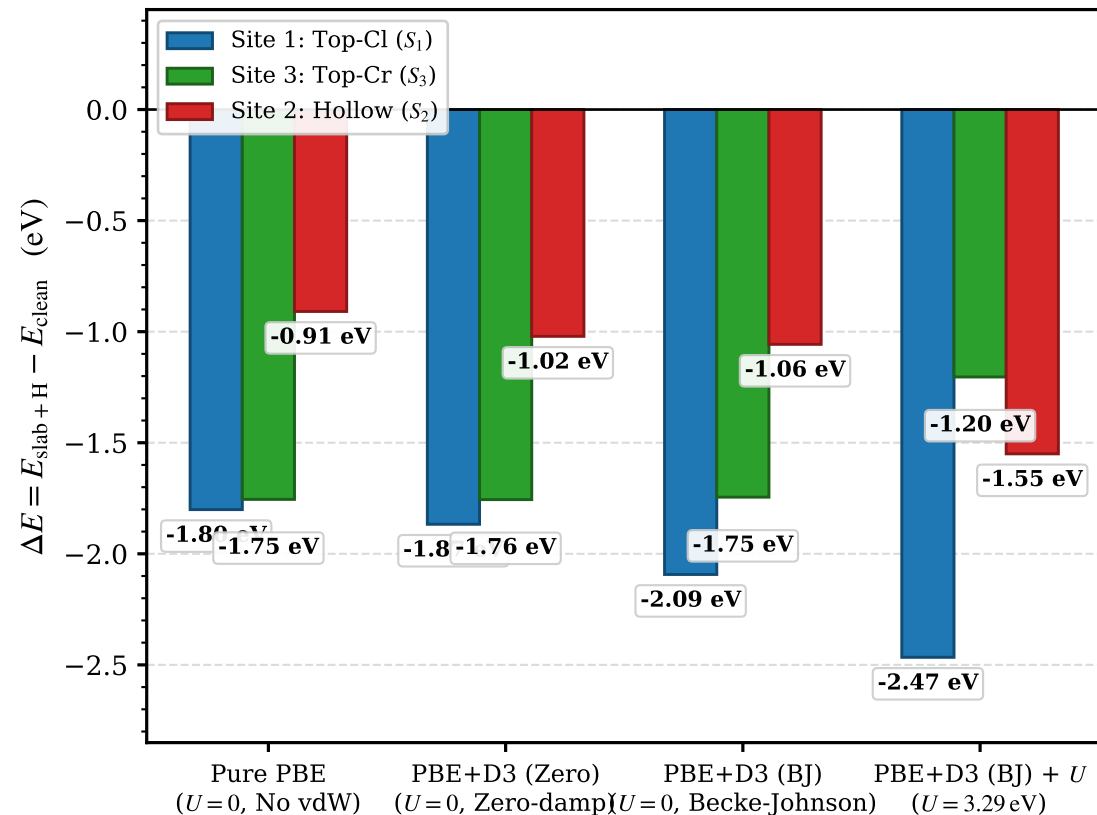


Comprehensive Hydrogen Adsorption Benchmark on 2×2 Monolayer CrCl_3 ($\theta = 0.25$, $d_{\text{H-H}} = 12.09 \text{ \AA}$)

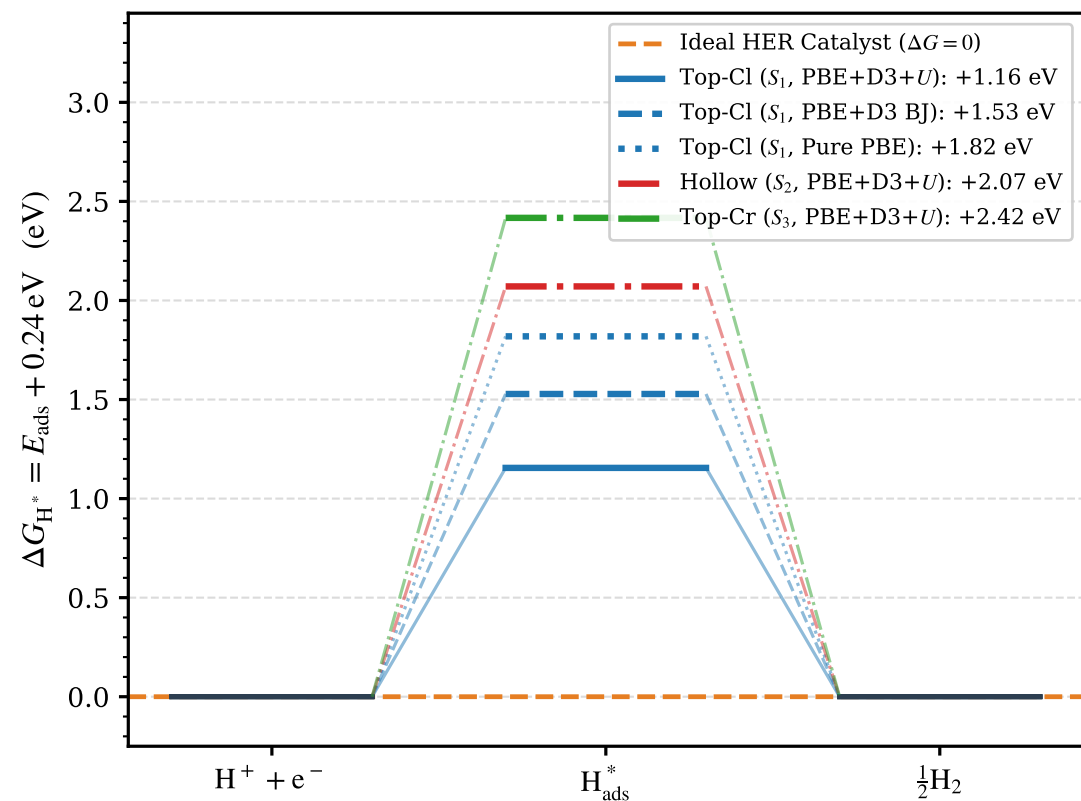
(a) 2×2 Hydrogen Adsorption Energy (E_{ads})



(b) Thermodynamic Binding Energy (ΔE)



(c) 2×2 HER Free Energy Profile ($\theta = 0.25$)



(d) Spin Polarization & Magnetic Compensation (2×2)

